

BACKLOG BUDDY ? STUDY MATERIAL

Material Science and Metallurgy ? Unit-wise Lecture Notes

Subject Code: ME303 Semester: 3

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Condensed notes covering all five units: Crystal Structure and Imperfections, Phase Diagrams, Heat Treatment of Steels, Mechanical Properties of Materials, Non-Ferrous Materials and Alloys.

Unit 1: Crystal Structure and Imperfections

- Definition, key concepts and significance of Crystal Structure and Imperfections
- Important diagrams / derivations / algorithms for Crystal Structure and Imperfections
- Solved examples and previous-year questions on Crystal Structure and Imperfections

Unit 2: Phase Diagrams

- Definition, key concepts and significance of Phase Diagrams
- Important diagrams / derivations / algorithms for Phase Diagrams
- Solved examples and previous-year questions on Phase Diagrams

Unit 3: Heat Treatment of Steels

- Definition, key concepts and significance of Heat Treatment of Steels
- Important diagrams / derivations / algorithms for Heat Treatment of Steels
- Solved examples and previous-year questions on Heat Treatment of Steels

Unit 4: Mechanical Properties of Materials

- Definition, key concepts and significance of Mechanical Properties of Materials
- Important diagrams / derivations / algorithms for Mechanical Properties of Materials
- Solved examples and previous-year questions on Mechanical Properties of Materials

Unit 5: Non-Ferrous Materials and Alloys

- Definition, key concepts and significance of Non-Ferrous Materials and Alloys
- Important diagrams / derivations / algorithms for Non-Ferrous Materials and Alloys
- Solved examples and previous-year questions on Non-Ferrous Materials and Alloys

Exam Tips:

- * Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- * Solve at least 3 previous year papers under timed conditions before the exam.
- * Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- * Revise definitions, diagrams and standard derivations ? they earn easy marks.
- * Practise numerical problems from every unit; papers usually include 2-3 compulsory

problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.